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ICE2CPT 2026 Program Committee

Dear Program Committee,

We are pleased to submit the manuscript titled *"Retry Amplification in Distributed Systems: A Systematic Analysis of Retry Policies and Their Role in Cascading Failures"* by Rishabh Mehan and Jasmit Kaur Saluja for consideration at ICE2CPT 2026.

The paper addresses an understudied problem in distributed systems engineering: the emergent behavior of retry policies across multi-tier service architectures. Retry guidance is written for a single caller communicating with a single callee, and we found no prior systematic study of what those policies do collectively once every tier in a call path applies them simultaneously.

The contributions are as follows:

1. We formalize the **Retry Amplification Factor (RAF)**, a metric quantifying how retry policies multiply request volume during partial failures, and show that amplification compounds exponentially with service depth.
2. We present an **empirical study of 200 open-source Python microservice projects**. Explicit retry logic is detected in 11.5%, and an audit of our own false negatives places true prevalence near 41%. Among the projects detected, 60.9% contain at least one configuration without backoff, and exactly one of 113 production configurations randomizes its delay after manual verification.
3. We provide **simulation evidence** ($n = 100$ trials per configuration) that under correlated failure, standard retry policies achieve 25% *lower* success rates than no-retry baselines.
4. We propose the **Adaptive Retry Budgeting (ARB)** algorithm, which incorporates back-pressure signals and dynamic budget adjustment, maintaining baseline success rates while limiting amplification to near 1.0×.

Two limitations are stated plainly in the manuscript. All 200 analyzed repositories are Python, so the figures characterize one ecosystem rather than the Go-dominated candidate pool; and the evaluation runs in simulation rather than production, which buys controlled comparison at the cost of fidelity.

The manuscript is original work, has not been published elsewhere, and is not under consideration at any other venue. Both authors have approved its submission. Thank you for your consideration.

Sincerely,

Rishabh Mehan
on behalf of both authors